INCH-POUND

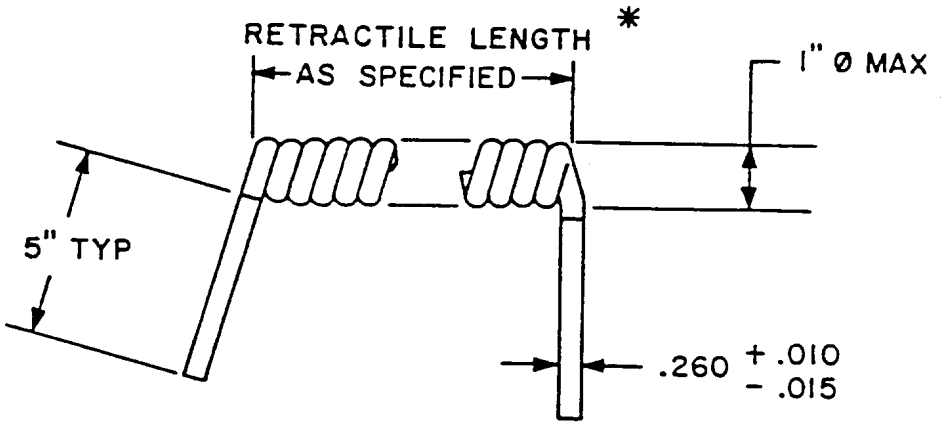
MIL-DTL-22442/3B(AS)
5 January 1998
SUPERSEDING
MIL-C-22442/3A(AS)
30 September 1991

DETAIL SPECIFICATION SHEET

CABLE ASSEMBLY, AIRCRAFT AUDIO, CABLE (RETRACTILE)

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-22442 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.



* ± 1" PER RETRACTED FOOT OR FRACTION THEREOF

FIGURE 1. P/N M22442/3.

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APPLICABLE DOCUMENTS:

MIL-I-3930 - Insulating and Jacketing Compounds, Electrical (for Cables, Cords and Wires), General Specification for

MIL-C-13273 - Cord, Electrical (Retractile 2, 3, and 4 Conductor WD-9/U, WT-2/U and WF-4/U

MIL-HDBK-454 - Electronic Equipment, General Guidelines for

REQUIREMENTS:

1. All performance requirements and construction requirements except as modified by this specification sheet shall be in accordance with MIL-C-13273.

* 2. Materials. All materials used in the construction of the cable assemblies should be fungus-resistant as specified in Guideline 4 of MIL-HDBK-454.

3. Conductors. Each stranded conductor shall be equivalent to 404 circular mils (No. 24 AWG) and shall consist of 4l strands of No. 40 AWG tinned soft cadmium copper alloy, having a l percent nominal content of cadmium. It shall be tight bunch stranded with a lay not exceeding 0.75 inch. Prior to stranding, the elongation of individual strands shall be not less than 10 percent and the tensile strength shall not be greater than 55,000 pounds per square inch. After stranding, and prior to insulating, the elongation of individual strands shall not be less than 5 percent and the tensile strength shall not be greater than 58,000 pounds per square inch. Each conductor in the finished cord shall have a nominal DC resistance of 39.0 ohms per 1000 feet at or corrected to a temperature of 20°C.

4. Insulation. The insulation for the retractile cords shall be Type IR or Type IH rubber compound conforming to the requirements of MIL-I-3930 and capable of meeting the requirements specified herein.

5. Dimensions of insulated conductors. The diameter of each insulated wire stranded conductors shall be not less than 0.057 inch nominal and shall not be greater than 0.063 inch. The minimum wall thickness shall not be less than 0.009 inch.

6. Colors. The color of the insulation on the 4 conductors shall be black, white, red and green.

* 7. Separators. The insulation on each conductor shall be free stripping. Separators shall consist of a serving, knitted barrier, or a close braid of suitable material. The use of a separator is optional.

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* 8. Stay-cord. The stay-cord shall have a minimum breaking strength of 25 pounds and shall not abrade or otherwise damage the insulation when the cord is bent or stretched.

* 9. Binder. Use of a binder is optional.

10. Shield. The shield shall consist of 32 nominal strands of No. 36 AWG tinned soft cadmium copper alloy having a l percent nominal content of cadmium. The shield shall be served over each insulated conductor in an even helix (laid adjacent to each other) with a 88 percent minimum coverage. A served shield shall be applied over each black, red and green insulated conductor separately.

* 11. Outer jacket. The cabled conductors shall be jacketed with type JS-L(SBR), or JH(EPDM) compound per MIL-I-3930 or a SBR natural rubber blend compound conforming to the performance requirements of MIL-C-l3273. The color of the jacket shall be black.

* l2. Jacketed cord. After jacketing, the average overall diameter for the 4 conductor cord (with a individual shield around three conductors) shall be .260 +.0l0 -.015 inch. The average diameter shall be taken as a mean of the maximum and minimum diameter of l0 measurements at various points along the cord. Diameters of retractile cords shall be taken along the straight portions of the cord at both ends. The jacket thickness at any cross section of the cord shall be .010 inch minimum. The jacket shall be free-stripping from the insulation. A separator may be used under the jacket.

* l3. Construction. The cable shall be constructed of 4 insulated conductors, 3 of which are individually shielded, and a stay-cord. Filler may be added if necessary to make a circular cross section. The insulated conductors shall be twisted with or around the stay- cord, and the lay of the conductors shall be 1.50 inch maximum. A binder or separator may be applied over the cabled conductors.

l4. Retractile. The jacketed electrical cord shall be formed in a helix with close adjacent turns. The helical cord shall have a straight length of cord on each end tangent to the helical section. The finished electrical cord shall have a left hand lay in its helical section and shall have a smooth and uniform jacket. The helix shall have an overall diameter of not more than 1.0 inches.

* l5. Quality assurance provisions. Sampling inspection and test procedures for the cord construction shall be as specified in MIL-C-13273.

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* NOTES:

* 1. The parts specified on this specification sheet are for use on completed and tested cable assemblies. Qualification of individual parts and components is not required.

CHANGES FROM PREVIOUS ISSUE: The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
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